

Mining Terminological Knowledge in Biomedical Text Abstracts

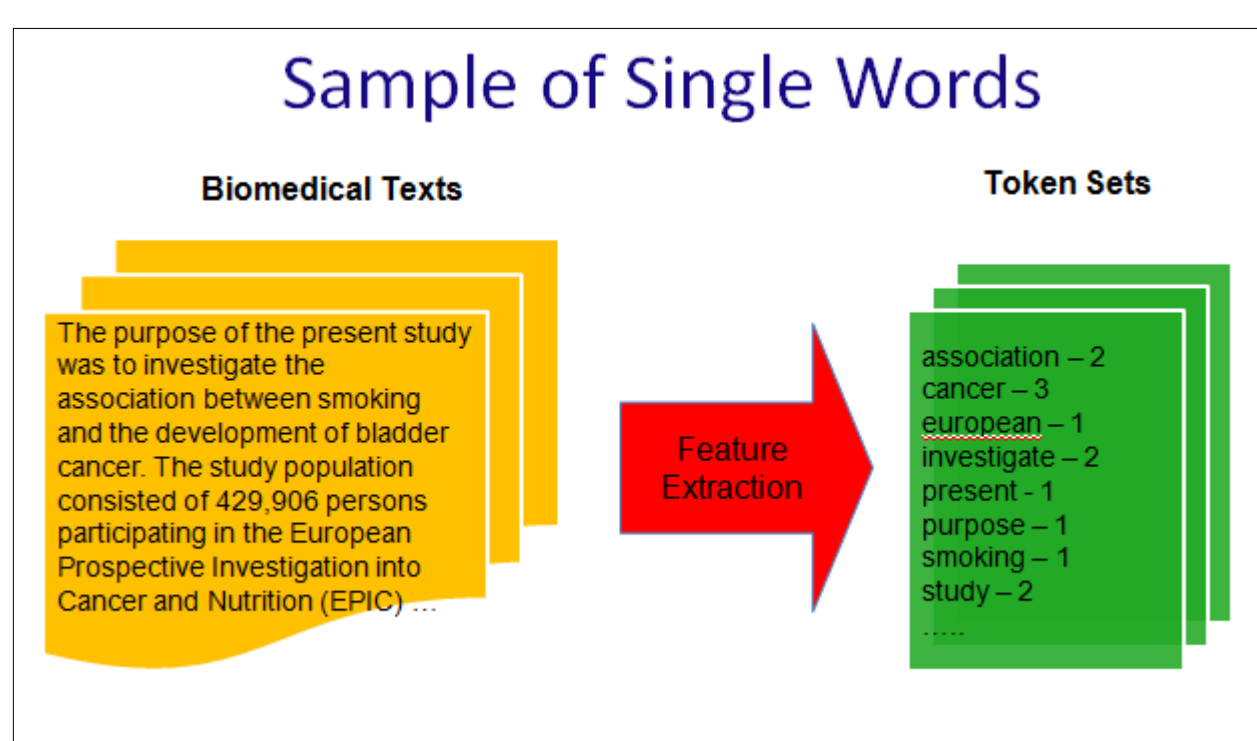
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Introduction

- ❖ Terminology management, especially term recognition is very important to extract relevant terms for searching relevant documents in database.
- ❖ Our main purpose is to investigate and evaluate the performance of different types of features for classifying biomedical text abstracts.
- ❖ Types of features;
 1. Single words
 2. Compound words
 3. Named entity
- ❖ We conduct the experiments by classifying biomedical text abstracts using different feature sets extracted from selected human diseases category.

Feature Extraction

- ❖ There are several approaches to recognize and extract features in biomedical literature.
- ❖ We apply three different approaches to extract different types of features from biomedical text abstracts.

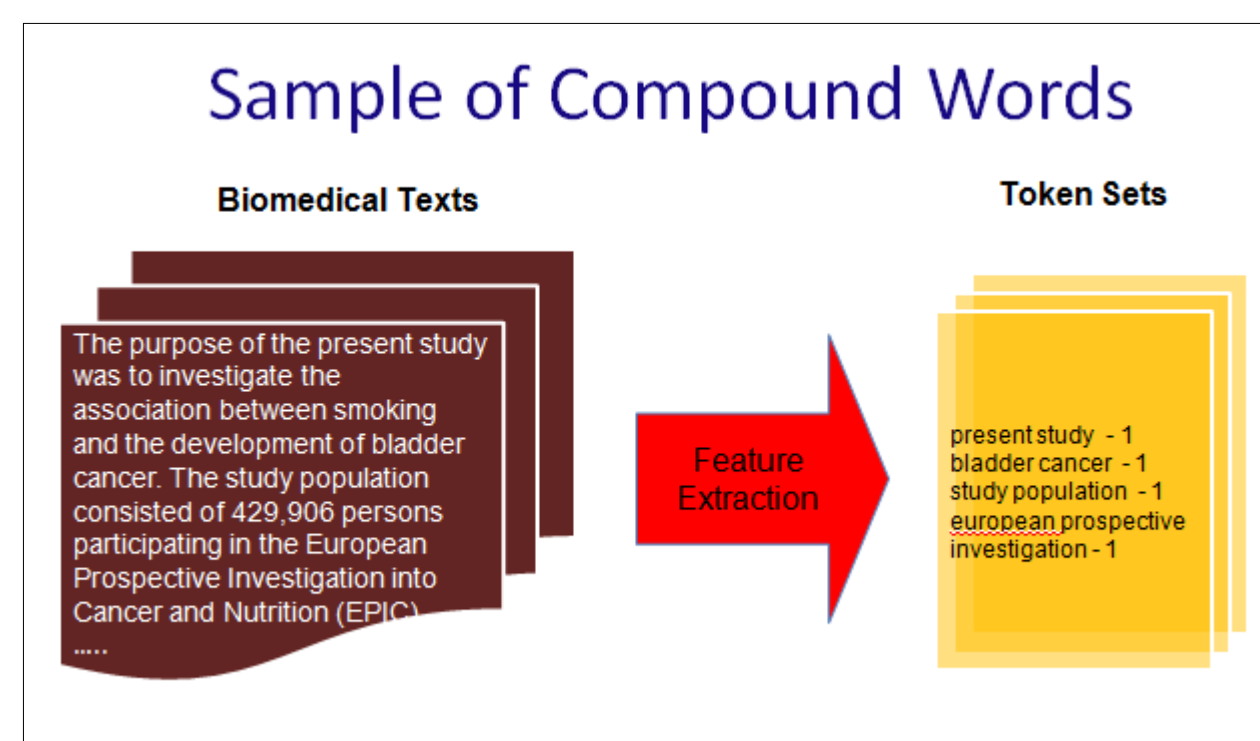


1 Single Word Approach

- ✓ Extract every single words (tokens).
- ✓ Remove common words using stop words list.
- ✓ Merge word variants and convert to canonical word.

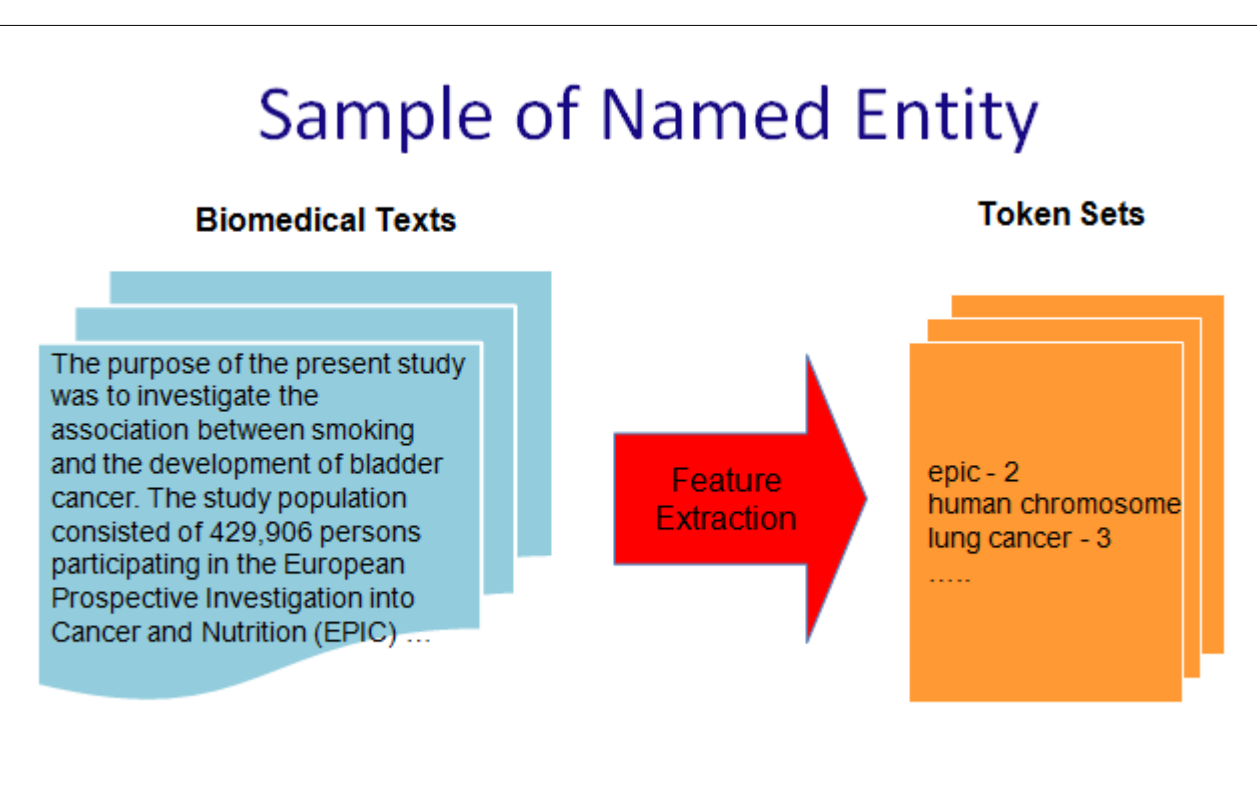
2 Compound Word Approach

- ✓ Extract terms using chunking method.
- ✓ Select noun phrases and verb phrases from chunking output.
- ✓ Convert to canonical word.



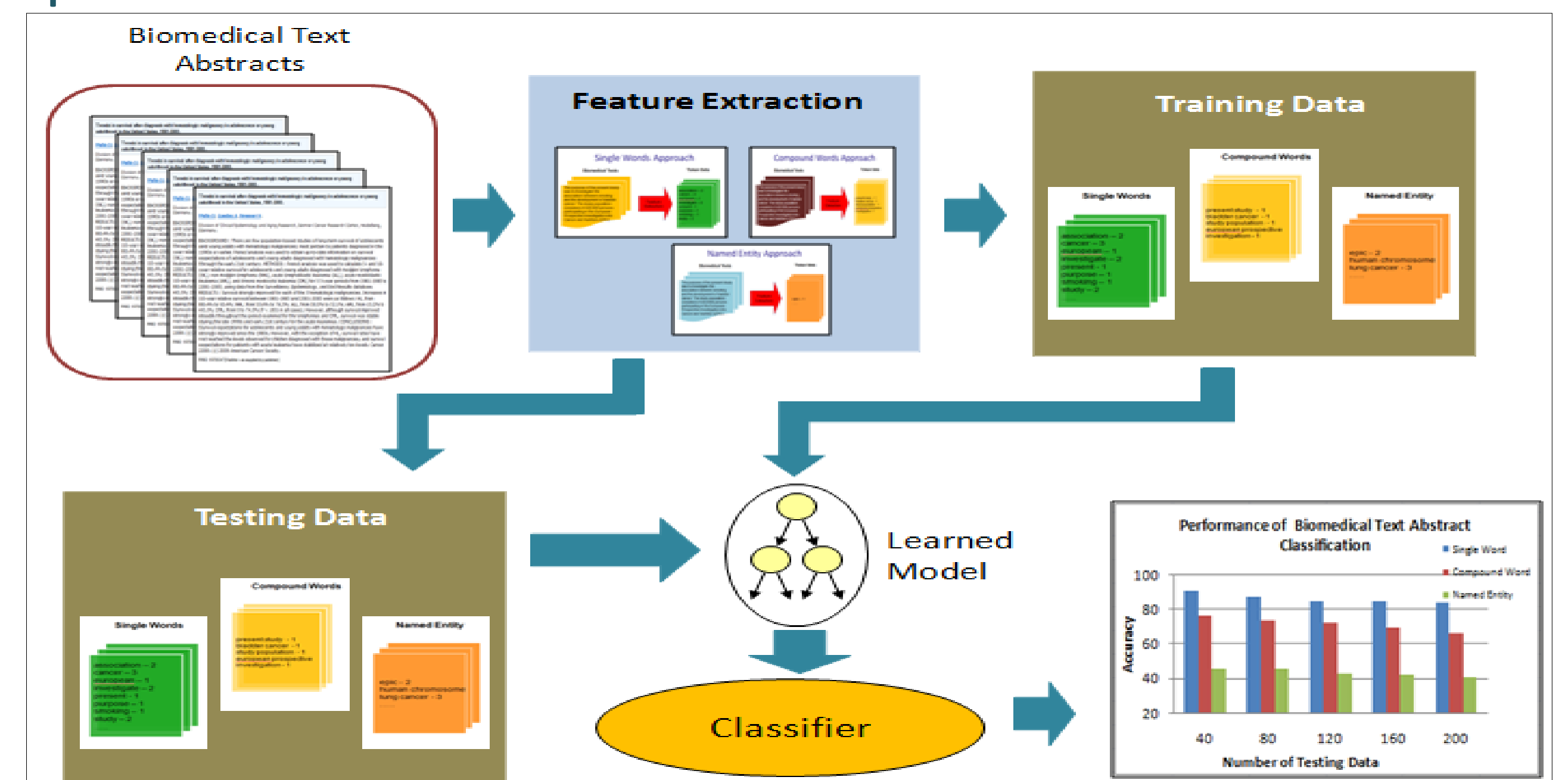
3 Named Entity Approach

- ✓ Extract named entity using named entity recognition tools.
- ✓ Select five entity types only, such as protein, DNA, RNA, cell type and cell line.



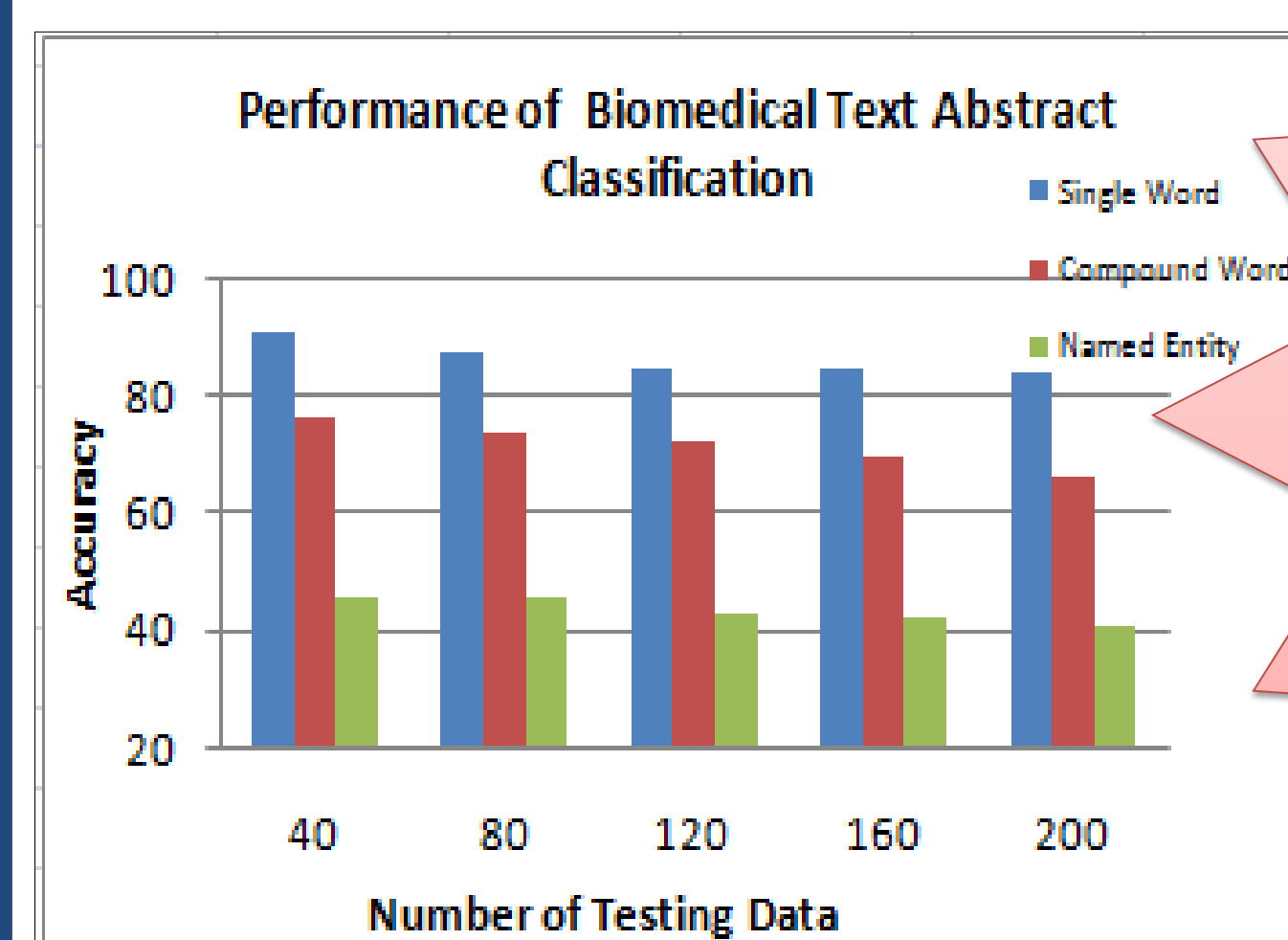
Methodology

- ❖ Feature extraction aim is to recognize and extract relevant features from biomedical text abstract.
- ❖ Then, selected features (as the text string representation) will be transform into numeric feature vectors, where we represent our documents as vector space model.



Experimental Results

- ❖ We experiment with 400 biomedical text abstracts that downloaded from Medline database.
- ❖ In this experiment, we extract three categories of feature sets using different term recognition approaches.
- ❖ Finally, we used these features sets for biomedical text classification experiments.



Results - different approaches of term recognition produce different performance of biomedical text classification.

Conclusions

- ❖ We evaluate the performance of using different term recognition approaches by extracting different types of features for classifying biomedical text abstracts.
- ❖ Overall, single words as features perform better than other category of features.
- ❖ Our future work is to increase the performance of biomedical text classification by extracting more relevant features, especially for compound words and name entity features.

Acknowledgements

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